

Sujit Medha Makkapati

Senior Data Engineer

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PROFESSIONAL SUMMARY

Results driven Senior Data Engineer with 4+ years of experience designing, building, and optimizing large scale data pipelines and cloud native data platforms across healthcare, manufacturing, and technology sectors. Passionate about turning raw data into reliable, scalable infrastructure that business teams can trust. Whether it is architecting a real time streaming pipeline on AWS Kinesis or designing a star schema warehouse in Snowflake, I bring a hands on approach and a strong sense of ownership to every project. Known for collaborating closely with data scientists, analysts, and business stakeholders to deliver solutions that are not only technically sound but genuinely useful. Equally comfortable leading complex migrations, mentoring junior engineers, or optimizing a slow dbt model at 11 PM before a product launch.

CORE COMPETENCIES & TECHNICAL SKILLS

Cloud Platforms	AWS (S3, Glue, Redshift, EMR, Lambda, Kinesis, Athena, EC2, IAM) · Azure (ADF, ADLS Gen2, Synapse, Databricks) · GCP (BigQuery, Dataflow, Pub/Sub, GCS)
Data Warehousing	Snowflake · Amazon Redshift · Azure Synapse Analytics · Google BigQuery · dbt (Core & Cloud) · Star Schema · Data Vault 2.0 · Kimball Methodology
Big Data / Spark	Apache Spark (PySpark & Scala) · Apache Kafka · Apache Flink · Delta Lake · Apache Hudi · Databricks · AWS EMR · HDFS · Hive · Presto
Orchestration / ETL	Apache Airflow · AWS Glue · Azure Data Factory · dbt · Prefect · Informatica · Talend · Luigi · Great Expectations
Programming	Python (Pandas, NumPy, SQLAlchemy, Boto3, PySpark) · SQL (Advanced) · Scala · Bash/Shell · Java (Basics)
Databases	PostgreSQL · MySQL · MongoDB · Cassandra · DynamoDB · Redis · Elasticsearch · Oracle · MS SQL Server
DevOps / MLOps	Docker · Kubernetes · Terraform · GitHub Actions · Jenkins · CI/CD · Helm · AWS CDK · Ansible
Data Quality / Gov.	Great Expectations · Monte Carlo · dbt Tests · Data Contracts · Apache Atlas · Collibra · AWS Glue Data Catalog
Visualization	Power BI · Tableau · Looker · QuickSight · Grafana · Matplotlib · Seaborn
Soft Skills	Cross - functional Collaboration · Stakeholder Communication · Agile / Scrum · Technical Mentoring · Root - Cause Analysis

PROFESSIONAL EXPERIENCE

Senior Data Engineer | SkyMindTech - US (Remote)

Feb 2025 - Present

- Architected and deployed a real time patient data ingestion platform on AWS using Kinesis Data Streams, Lambda, and Glue, reducing data latency from 4 hours to under 5 minutes and enabling same day clinical decision support.

- Designed and implemented a multi-layer data lakehouse on AWS S3 with Delta Lake (bronze / silver / gold zones), processing 2 TB+ daily and serving as the single source of truth for 15+ downstream analytics and ML applications.
- Built 80+ modular dbt models in Snowflake following Kimball dimensional modeling principles, cutting average dashboard query time by 68% and eliminating 12 redundant manual reporting processes.
- Led migration of legacy on premise Oracle ETL pipelines to a fully managed AWS Glue + Airflow architecture, reducing infrastructure costs by 35% and achieving 99.95% pipeline uptime in the first quarter post migration.
- Implemented a comprehensive data quality framework using Great Expectations integrated into CI/CD pipelines, enforcing schema validation, null checks, and statistical anomaly detection across 40+ critical data assets.
- Collaborated with ML engineers to build feature engineering pipelines in PySpark on Databricks, reducing feature computation time by 55% and supporting the delivery of three production ML models.
- Established data governance standards including metadata cataloging (AWS Glue Catalog), PII masking with AWS Macie, and column level security in Snowflake to meet PIPEDA and HIPAA compliance requirements.
- Mentored 3 junior data engineers through code reviews, pair programming, and internal documentation, raising the team's average PR review cycle from 3 days to under 1 day.
- Defined and monitored SLA metrics for 50+ Airflow DAGs using CloudWatch dashboards and PagerDuty alerting, maintaining an average pipeline success rate of 99.8%.

Data Engineer | Toshiba Global Commerce Solutions - US (Hybrid) Jun 2023 - Jun 2024

- Designed and maintained end to end ETL pipelines on Azure Data Factory to ingest IoT sensor data from 3,000+ POS devices across 200 retail locations into Azure Data Lake Storage Gen2, processing 500M+ events per day.
- Built a streaming analytics solution using Azure Event Hubs and Apache Kafka to detect real - time anomalies in transaction data, reducing fraud detection lag from T+1 day to under 30 seconds.
- Developed PySpark jobs on Azure Databricks to transform and aggregate raw transactional data into a star schema warehouse in Azure Synapse Analytics, enabling self - service reporting for 50+ business users.
- Optimized Synapse SQL pools by redesigning table distribution strategies (hash vs round robin), implementing result set caching, and partitioning large fact tables, improving report query performance by an average of 72%.
- Created and maintained 60+ dbt models to implement business logic transformations, enforce data contracts, and generate automated data lineage documentation reviewed during quarterly audits.
- Automated infrastructure provisioning using Terraform and Azure Bicep for Databricks clusters, ADF pipelines, and ADLS containers, reducing environment setup time from 2 days to under 1 hour.
- Integrated Power BI dashboards with Synapse Analytics using DirectQuery and scheduled import modes, delivering real time inventory and sales insights that reduced manual reporting effort by 20 hours per week.
- Participated in Agile sprints using Jira and Confluence, delivered 100% of assigned story points across 8 consecutive sprints, and authored technical runbooks adopted as team standards.

Junior Data Engineer | DhyanaHitha Technologies - Hyderabad, India Jun 2021 - Nov 2022

Indian tech services firm delivering data solutions for healthcare and education clients.

- Built and maintained Python based ETL pipelines using Pandas and SQLAlchemy to extract data from 10+ heterogeneous sources (REST APIs, SFTP, MySQL, Oracle) and load into a centralized PostgreSQL data warehouse.
- Developed Apache Airflow DAGs to orchestrate complex multi-step workflows with SLA - based alerting, replacing cron jobs and reducing pipeline failures by 60% within the first 3 months.
- Wrote advanced SQL queries and stored procedures in PostgreSQL and MySQL for data aggregation, generating weekly KPI reports consumed by 5 client organizations.
- Assisted in designing a dimensional data model (fact and dimension tables) on AWS Redshift for an Elearning analytics platform, supporting 1M+ student records and enabling cohort analysis.
- Implemented basic data quality checks using Python scripts to validate row counts, referential integrity, and business rule compliance before daily data loads, reducing downstream reporting errors by 45%.

- Containerized ETL applications using Docker and managed deployments on AWS EC2, reducing environment inconsistency issues and enabling reproducible builds across dev, staging, and production.
- Created Tableau dashboards for a healthcare client to visualize patient admission trends, appointment utilization rates, and bed occupancy metrics, supporting operational planning for a 500 bed hospital.

EDUCATION

Master of Science - Computer Science | University of North Texas

Denton, TX

Relevant Coursework: Distributed Systems, Advanced Database Systems, Big Data Analytics, Machine Learning, Cloud Computing

Bachelor of Technology - Computer Science & Engineering | GITAM University Hyderabad, India

KEY PROJECTS & HIGHLIGHTS

Real - Time Healthcare Data Platform (SkyMindTech)

- End to end streaming + batch lakehouse: Kinesis - Lambda - S3 (Delta) - Snowflake - Power BI; 2 TB/day throughput; 99.95% uptime SLA.

Retail IoT Anomaly Detection System (Toshiba)

- Kafka + Event Hubs streaming pipeline detecting POS transaction anomalies in < 30 seconds across 200 stores; reduced fraud exposure by 18%.

Healthcare KPI Data Warehouse (Dhyanahitha Technologies)

- Designed and built PostgreSQL + AWS Redshift dimensional model ingesting 10 source systems; automated via Airflow; cut reporting errors by 45%.
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